HAEBOT Hand Controller v2 — what each chip does

115 × 56 mm custom board, Raspberry Pi removed. Real product photos of each part, with a one-line explanation you can use when a senior asks.

Motor connector (XH-4)
4-pin plug for the NEMA 17 base motor coils (A1 A2 B1 B2).

Servo headers (3×8)
6 MG996R plugs in here: fingers on CH0–4, wrist on CH5, two spares.



TMC2209
Quiet stepper driver that spins the NEMA 17 base motor.



PCA9685
16-channel servo controller; makes the PWM signals for the finger and wrist servos.



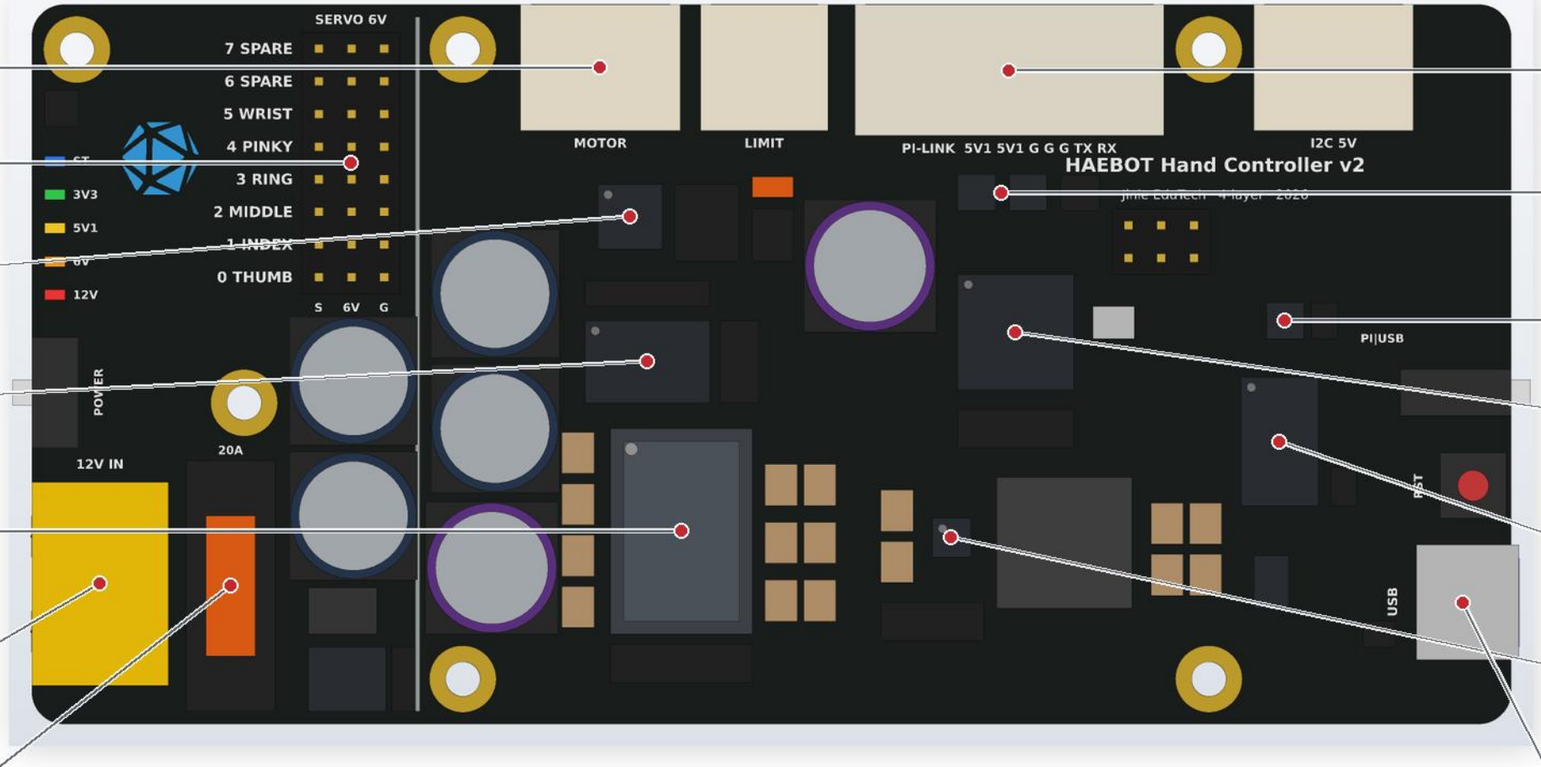
TPSM41625
Power module that turns 12V into 6V (up to 25A) for all 6 MG996R servos.



XT60 12V input
Locking 12V inlet from the Mean Well supply (polarized, 30 A rated).



20 A mini blade fuse
Main fuse on the 12V input; blows before the power supply overloads.





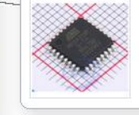
Molex Micro-Fit 3.0 7-pin
Locking cable connector carrying 5V power + UART to the Raspberry Pi.



SN74LVC1T45 (×2)
Level shifter so the 3.3V Pi and 5V ATmega can talk over UART safely.



TLV75533P
Small regulator making clean 3.3V for the PCA9685 and logic.



ATmega328P
The main microcontroller (same brain as an Arduino Nano) running the hand firmware.



CH340C
USB-to-serial chip so the PC can program/talk to the ATmega over USB-C.



TPS56837
Turns 12V into 5.1V (up to 8A) to power the Raspberry Pi and logic.



USB-C connector
Programming / debug port for the ATmega (firmware is uploaded over USB-C).



Mean Well LRS-350-12
Off-board power supply: wall AC to 12V

Sits next to the robot, not on this board. A 12 V, 29 A (350 W) enclosed supply. We run it well below its limit (about 11 A / 135 W in the worst case).